

SQL FUNDAMENTALS

EXERCISE 2

SQL AGGREGATE FUNCTIONS AND OPERATORS ON THE STUDENT TABLE

TABLE - student

STUDENT-ID	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	24	HR

Q1: List all distinct departments in the students table.

Answer: `SELECT DISTINCT department
FROM students;`

Table: student

department
HR
IT
Finance

Q2: Get the average age of students per department

Answer: `SELECT department, AVG(age) AS avg-age
FROM students
GROUP BY department;`

department	avg age
IT	20.5
HR	23
Finance	23

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Q3: Show departments with more than one student.

ANSWER:
 SELECT department,
 COUNT(*) AS count_students
 FROM student
 GROUP BY
 department
 HAVING
 COUNT(*) > 1;

Table: student

department	count_students
IT	2
HR	2

Q4: Get all students whose age is between 21 and 23

ANSWER:
 SELECT *
 FROM student
 WHERE age BETWEEN 21 AND 23;

Table: student

student_id	name	age	department
2	Bob	20	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	20	HR

Q5: List all students in the IT or HR department who are older than 21.

Answer:

```
SELECT *
FROM student
WHERE (department = 'HR' OR department = 'IT')
AND age > 21;
```

Table - student

Student_id	Name	age	department
2	Bob	22	HR
5	Eve	23	HR

~~CREATE COURSES TABLE~~

Table: courses

Course-id	course-name	department	Credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

Q6: Show total credits per department, only for departments with more than 5 total credits.

```
Answer: SELECT department,
SUM(Credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(Credits) > 5;
```

Table: courses

department	total_credits
IT	8

Q7: List all courses that do not have 4 credits

Answer:
 SELECT *
 FROM courses
 WHERE credits <> 4;

Table: courses

course_id	course_name	department	Credits
101	Basic SQL	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8: Show top 3 courses by credits in DESC order

Answer:
 SELECT
 course_id, course_name, Credits
 FROM courses
 ORDER BY credits DESC, course_name ASC
 LIMIT 3;

Table: courses.

course_id	course_name	credits
103	Data Science	4
102	Python	4
101	SQL Basics	3.

Q9:

03: The ENROLMENTS TABLE

Table: enrolments

enrollment_id	Student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90 max
4	4	104	88
5	5	105	82 min

Q9. Get the maximum, minimum, and average grade across all enrolments

Answer:

```
SELECT
    MAX(grade) AS max-grade,
    MIN(grade) AS min-grade,
    AVG(grade) AS avg-grade
FROM
    enrolments;
```

Table: enrolments

max-grade	min-grade	avg-grade
90	82	84.6

Q10. Count how many enrolments exist per Course

Answer:

```
SELECT
    course_id,
    COUNT(*) AS enrolment-count
FROM
    enrolments
GROUP BY
    course_id;
```

Table: enrolments

enrollment_id	enrollment-count
1	1
2	1
3	1
4	1
5	1

Q4: THE SALARIES TABLE

TABLE: Salaries

employee_id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	FINANCE	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

Q11: Find total salary and total bonus per department.

Answer:

SELECT

department,
SUM(salary) AS total_salary
SUM(bonus) AS total_bonus

FROM

salaries

GROUP BY

department;

TABLE: Salaries

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
FINANCE	70 000	6 000

Q 12: Show departments where average salary is above 55000

ANSWER:

SELECT

department,
AVG (salary) AS avg-salary

FROM

Salaries

GROUP BY

department

HAVING

AVG(salary) > 55000;

Table: salaries

department	avg-salary
IT	61000
HR	Not included 54500
Finance	70000

Q 13: List employees whose salary plus bonus is greater than 60000.

ANSWER: SELECT employees.id, name, salary, bonus,
(salary + bonus) AS total-compensation
FROM Salaries
WHERE (salary + bonus) > 60000;

Table: salaries

employees.id	name	Salary	bonus	total compen
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

Q5: The projects Table

TABLE: Projects

Project-id	Project name	department	budget
1	AI APP	IT	120 000 ✓
2	Payroll System	Finance	8000 ✓
3	Dashboard	IT	150 000 ✓
4	Website	Marketing	60 000
5	HR Portal	HR	50 000 ✓

Q14: Show total and average budget per department.
Only include departments with average budget above 70 000.

ANSWER: SELECT department,
SUM(budget) AS total-budget,
AVG(budget) AS avg-budget
FROM projects
GROUP BY departments
HAVING AVG(budget) > 70 000.

TABLE: Projects

department	total budget	avg budget
IT	270 000	135 000

Q15: List all projects with budgets between 50 000 and 100 000, excluding the marketing department.

ANSWER: SELECT *
FROM projects
WHERE budget BETWEEN 50 000 AND 100 000
AND department <> 'Marketing';

TABLE: Projects

Project-id	Project name	department	budget
1	AI APP	IT	120 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000